

History and philosophy of geography, 2002–2003: geography in its place

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I Geographies of knowledge and of science

'A glance out across fields of knowledge quickly reveals that "space" became the master metaphor of late twentieth-century epistemology.' So Daniel Brewer, historian of Enlightenment France, begins his summary discussion of the notion of space in the eighteenth century (Brewer, 2004: 171). For Brewer, the importance of space rests not just in the fact that it has become the focus of intense theoretical reflection in a variety of disciplines (albeit that he pays geography the merest of sideways glances). Rather, space provides a way to understand the disciplinary production of knowledge in general and that to consider space as an interpretive practice – thinking geographically if you will – helps turn space into a visible object. Brewer focuses on 'physical space' (in looking at mapping), at 'social space' and 'colonized space', at 'epistemological space', and at 'esthetic' space. Brewer's concerns have been echoed in others' work. Much of it has focused on the eighteenth century, my second theme in this summary review. My third and final themes concern, respectively, mapping and questions of discipline and discourse (cf. Ryan, 2004). As will be apparent, these are divisions of convenience not mutually discrete categories. Let me begin by returning

to the centrality of geography to science and its spatial expression.

Questions concerning the geographical location of science and its movement from place to place are central to David Livingstone's concise yet rich survey *Putting science in its place*, subtitled *geographies of scientific knowledge* (2003a). The book is organized around three principal themes: the placed nature of science with reference to 'site' (the venues of science's making and reception), 'region' (different cultures of science) and 'circulation' (the movements of science). Livingstone's geographical and temporal sweep is wide and impressive: sixteenth-century domestic laboratories, seventeenth-century maps, Royal courts, botanical gardens, ships, the nineteenth-century archive, twentieth-century museums and much else. There is little, relatively, about the science of geography, for this is rather an account of how scientific knowledge in one guise or another has been differently made and differently received in different places. Space, questions of geography, are dealt with in much more than the trivial, the 'merely' ontological sense of location and distribution – after all, things do have to be somewhere. Livingstone's emphasis is on the ways in which questions of what is held to be science, or rationality, are always situated and always embodied, and

that questions of space and place make a difference to such matters. 'In different arenas different repertoires of practical rationality have been in operation, and forms of explanation, modes of practice, methods of justification, and traditions of inquiry central to one arena have been outlawed or marginalized in others. The *where* of scientific endeavour thus insinuates itself into science at all these levels' (Livingstone, 2003a: 180–81). These concerns will be familiar to geographers and others interested in the 'spatial turn' in science. But one has the feeling that the arguments informing this work of synthesis, rightly perhaps, are aimed more at historians (like Brewer) and other disciplines than they are at geographers alone.

To judge from at least three edited collections, Livingstone's message is being heard and shared. In *Instruments, travel and science* (Bourguet *et al.*, 2002), the focus is upon the travelling nature of knowledge – the central role of geographical displacement in science's making – and the contingent and located nature of instrumentality and precision. Noteworthy chapters include the geography of precision with respect to trade with Africa's Gold Coast (Schaffer, 2002), work on natural history, travel and artistic depiction (Bourguet, 2002), the sea trials of Harrison's chronometer between Jamaica and Portsmouth (Bennett, 2002) and Indo-British mapping practices in Central Asia (Raj, 2002). The studies making up *Travels of learning* (Simões *et al.*, 2003) are more geographically circumscribed, being concerned with the movement of science and of scientists as travellers within and from Europe's 'peripheries' with case studies from Portugal, Spain, Greece, Turkey, Russia, Hungary and the Scandinavian countries. The chapters contained in *Science and the city* (Dierig *et al.*, 2002a) are more local still, with studies of Dresden, Canton and, in more than one context, Paris figuring largely. In their introduction, the editors discuss the case for an 'urban history of science', a term they see uniting urban history and the history of science (Dierig *et al.*,

2002b: 1). Their work would be as well if not better described as an urban geography, even an urban historical geography of science, and would have as much unifying potential for the fields in question.

Individual studies elsewhere elaborate upon the importance of the local context in the geographies of knowledge. For Matless in his work on ecological science in the Norfolk Broads (2003), what counted as local knowledge was neither straightforward, nor readily accepted by the 'scientific community' and the complex relationship between such terms makes their simple separate definition problematic. Others have turned their attention to fieldwork and to the geographies of ecological and civic science at local, regional, national and imperial scales and to the relationship between them (Anker, 2001; Dominy, 2002; Cameron and Matless, 2003; Lachmund, 2003; Naylor, 2003; Withers and Finnegan, 2003). The question of 'national science' has been called into particular question by one leading historian of science who trenchantly dismisses the term in favour of 'an enlarged perspective focusing on regional manifestations of transnational practice' (Pyenson, 2002: 253). It is unlikely that 'the national' will be readily dismissed as a scale of inquiry, but what this and other claims signal to is a continued interdisciplinary engagement with questions of space, place and movement and with the language of geography and geographical scale in explaining the nature of knowledge. Nowhere is this more apparent perhaps than in recent work on the eighteenth century.

II Eighteenth-century worlds

Scholars of the eighteenth century and historians of ideas in general have paid fulsome tribute to Roy Porter, whose death on 3 March 2002 robbed the world of letters of a remarkable and industrious intellect. In a theme issue of *History of Science*, Porter's influence in the history of medicine and psychiatry (Andrews, 2003), the earth sciences (Rudwick, 2003) and Enlightenment studies

(Schaffer, 2003), among other topics, is acknowledged. Porter's last two books exemplify his range and incorporate geographical matters in different ways. His *Flesh in the Age of Reason* considers the discursive and physical constitution of the human body and the relationships between mind and body in historical context. Industrial bodies, sick bodies, pictured bodies: here is a man of science in high command of the 'Science of Man' (Porter, 2003a).

The 36 essays brought together in Porter's last edited work paint an even wider canvas (Porter, 2003b). Space precludes their detailed examination but two points may be made of the collection so far as it relates to geography. Disappointingly, there is no formal explicit attention to geography, an omission explained as a restriction upon space. Yet the volume is suffused with attention to the geographies of science in ways signalled to above: to the sites of science's making, to the varying geographical reception of Newtonianism, for instance, to oceanic navigation, and in the attention to nonwestern traditions. In effect, this is a topographic map to the eighteenth-century sciences. The essays might be usefully read in conjunction with the many shorter articles making up the four-volume *Encyclopedia of the Enlightenment* (Kors, 2003). Two other edited works offer a more consciously British (not to say London-based) view of the Enlightenment world, a metropolitan calculus that marks the 250th anniversary of the establishment of the British Museum (Anderson *et al.*, 2003; Sloan, 2003).

Geography as a subject in the eighteenth century has been examined (Livingstone, 2003b; Withers, 2003). From an edition of de Saussure's lectures on physical geography given in Geneva in 1775, we are afforded insight into what was being taught in at least one place. The editors additionally discuss the evolution of the term 'géographie physique' from the eighteenth century onwards (Carozzi and Newman, 2003). Smith and Vining (2003) provide a prosopographical study of the

geographical community in the early American Republic. Altogether grander surveys of the geographies of the eighteenth-century world are provided by Colley (2003) in her *Captives: Britain, empire and the world 1600–1850*, and by the chapters in Nussbaum's *The global eighteenth century* (2003). Colley's book is grand-sweep narrative history from the bottom up, offering stories of empire not as an unproblematic record of near-global governance but of individuals' lives, single narratives played out against the 'smallness' of Britain and the vastness of its imperial enterprise. It is, moreover, acutely sensitive to the geographical diversity of the British empire.

Two detailed studies offer different perspectives on the place of geography as eighteenth-century global measurement. Before the late 1730s or so, no one could agree what shape the world was, for, depending on whose gravitational theories one subscribed to, Newton's or Descartes', the globe was a different shape: melon or pumpkin. The resolution of these competing theories (in Newton's favour – the Earth is pumpkin-shaped, flattened at its poles) via the expeditions of La Condamine to Peru and Ecuador and of Maupertuis to Lapland is documented in Terrall's fine biography of the latter figure (2003). Alder's focus is on the less successful expeditionary endeavours of two other Frenchmen, Delambre and Méchain, whose task it was for seven years from 1792 to standardize the metre and to metricize the globe (Alder, 2002). Notably by persons encountering the Pacific region on behalf of European science and empires, geographies of measurement were also directed at people. It was in this new 'fifth division of the world' that anthropology and human geography have particular origins (Ryan, 2002). Unlike terrestrial measurement, ethnographic measurement was seaborne, islands points of refreshment, transient sexual liaison and sites of instrumental and human calibration, the peoples encountered the archetypes of invented racial typologies (Douglas, 2003; Rainbird, 2003;

Tcherkézoff, 2003). Nicholas Thomas adds to his other work on these questions – about the ambivalence of Pacific discovery, the faltering nature of empirical science at the edge of empire – in his *Discoveries* (2003). For Gascoigne, Australia ‘enters’ the world map, hesitantly shaped by Flinders’ 1801–1803 *Investigator* expedition, as an object of moral and natural improvement (2002). At much the same time, the world’s ‘fourth division’, the Americas, was being put to shape – by the Russians and, from 1803, by Americans themselves thanks to the Louisiana Purchase. Smith’s attention to the expeditions of the Russian geodesist Mikhail Gvozdev could be usefully read in conjunction with Frost’s reassessment of the Russian-employed Dane, Vitus Bering (Smith, 2002; Frost, 2003). Bering’s Great Kamchatka Expeditions to discover America via Siberia were motivated by commerce as much as by the need to get right the geography of the Strait that bears his name. In introducing a theme issue of *Historical Geography* on the Louisiana Purchase, Colten and DeLyser (2003) emphasize just how much the early United States took on – as new land, as a space of governance, as an imperial colony of alien peoples.

This and other work is illustrative of rich seams of scholarly inquiry. It is notable, however, that this is work examining the practices of geographical knowledge as eighteenth-century contemporaries understood them rather more than it is work disclosing the nature of eighteenth-century geography *per se*, or who then called themselves ‘geographer’, notwithstanding those few studies to the contrary noted above. If this is to note with others a difference between ‘discipline’ and ‘discourse’ (Livingstone, 2003b: 12–13; Mayhew, 2001; Ryan, 2004), it is not to see the distinction as securely fixed.

III Mappings

As Williams shows in his informative exegesis of the search for the Northwest Passage in the eighteenth century, mapping knowledge about the globe did not always translate into

knowledge-based maps. Geographical endeavour was delusional, mappable truth elusive in a region where the very indeterminacy of what was land and what sea or ice could make the difference between life or death (Williams, 2002). Neal makes just this point of Henry Briggs’s Northwest voyage in 1631, a voyage connected like its later counterparts with colonization and what she terms an emergent ‘navigational culture’ (Neal, 2002: 436). The discursive connection between geography and early modern mathematics was central to men like Briggs, as it had been to his predecessors such as Thomas Harriot and Martin Frobisher (Fox, 2001; McDermott, 2001). Securing one’s bearings were terrestrial and celestial matters. As Henry shows for Copernicus and for Francis Bacon and Purnfrey of William Gilbert, the Elizabethan ‘discoverer’ of latitude, these were revolutionary times in the sciences in which geography as a form of knowledge was centrally located (Henry, 2001; 2002; Purnfrey, 2002).

Maps do not reflect what is there so much as they reflect the concerns of their makers. Two studies look, for different periods, at the role of ‘cartographic fictions’ in America, the mismapping of others in popular culture and the errors that were unchallenged in maps (Piper, 2002; Schwartz, 2003). The issue of mapping people, ‘fixing’ them in the ‘empty’ spaces of the emergent United States, is part of the focus of Olund (2002) and Morin (2002). Here, the insights (and problems) of postcolonial theory are brought to bear in explaining the discursive production of ‘Native Americans’ and the politics behind ‘governable space’ in the later nineteenth century (see also Anderson and Domosh, 2002).

The power of maps as portable forms of geographical possession and representation is clear in several studies of the precolonial and early colonial worlds. Issues to do with the map as, for example, a form of mobile immutable authority, of other’s rule over space, are evident in early modern Japan

(Yonemoto, 2003), colonial America (Pritchard and Taliaferro, 2002; MacMillan, 2003), and Renaissance Europe (Buisseret, 2003). What connects these studies are ideas about maps as particular forms of what Shalev (2003) terms 'visual erudition' and what Buisseret terms a broader visual thinking of which 'map' or 'cartographic consciousness' was part. There are connections, too, with the ideas reviewed above in regard to where such 'visual erudition' or 'cartographic consciousness' was acquired: for Shalev via bible reading, or in institutions where the claims of some mapmakers could be taken as more trustworthy and 'accurate' than others (Withers, 2002). Surveying was a form of mapping and of imperial viewing which had to be learnt as well as technically accomplished 'out there', whether the 'field' was North India (Raj, 2002), Ceylon (Barrow, 2003) or in the offices of the Royal Geographical Society (Collier and Inkpen, 2003). Whether for botanical geography (Rupke, 2001) or for the mapping of science in the nineteenth-century city (Picon, 2003), maps provide a powerful means of spatializing knowledge.

IV Discipline and discourse

Historians of geography looking for the subject on the map of the nineteenth-century human and social sciences provided by Theodore Porter (2003) will be disappointed. Only two pages are devoted to human geography. This paucity of treatment is unfortunate for two reasons. First, as is well understood, the later nineteenth century witnessed the institutionalization and disciplinary professionalization of geography in its 'modern' form, a history which has been examined in different national context by the studies in Dunbar (2001: see also Michie and Thomas, 2002). Secondly, geography's modern making, then as now, was bound up with wider epistemological debates: who was credible and how could we trust their testimony (Harris, 2002; Kusch and Lipton, 2002), what was a 'fact' in historical inquiry

(Torstendahl, 2003) and who in science could be deemed professional by virtue of their language (Barton, 2003).

A more detailed guide to geography's place in one national context is provided by the 20 chapters making up Johnston and Williams' *A century of British geography* (2003). The century in question is defined by the centenary of the British Academy despite, as the editors point out, that body having figured only lightly in the institutional record of British geography's professionalization and promotion. For that reason, perhaps, the collected essays are not underscored by a self-indulgent sense of institutional commemoration. Rather, their concern is to review geography's place in helping promote an informed citizenry (which I am not always sure it has done despite claims now and then to the contrary) and to record what we might think of as subdisciplinary accomplishments (which, overall, the essays do well). After two introductory essays (Johnston, 2003a; Livingstone, 2003b), the book has five main headings: 'Environment', 'Place', 'Space', 'Geography in action' and 'Geography moving forwards'. There is not the space here to discuss individual themes and chapters, but two more general issues strike me. First, the various themes afford one opportunity – to be used in teaching perhaps? – to compare similar 'moments', several of which are noted in the bibliographies, in which the state of the discipline has been reviewed. What has been the language, the range, the basis to professional credibility in British geography in (say) 1951, in the 1970s, in the 1890s? Secondly, it is far from clear whether this is a record of British geography or of geography in Britain. The difference might matter. For, while we are given partial insights into shared concerns that transcend national boundaries in ways Pyenson (2002) would applaud, it is also unclear how what we are told complements other 'national' traditions, or accords with other ways of telling geography's story. Johnston makes just such a point in considering the history of geography's

'quantitative revolution' in the late 1960s and 1970s (Johnston, 2003b). Keylock (2003) reflects on this for physical geography.

How we recount geography's history and for whom are among the questions raised by Lorimer's attention to 'geography's small stories' (Lorimer, 2003a; 2003b). In differently experienced versions of the field excursions, Lorimer traces accounts of 'doing geography' as a lived practice differently recorded – on maps, in diaries, in photographs. In being informed by ideas in science studies on the sites of knowledge's making, this work returns us to the connections between what geography is held to be and the spaces in which it was made. Others have posed similar questions of public statistics, polar exploration, evolutionary biology and oceanography. (Schwartzman, 1999; Wylie, 2002; Elliott, 2003; Rainger, 2003). In other work, the personalities behind earlier forms of human geography have been documented for America, Finland and France (Koelsch, 2003; Clout, 2003a; 2003b; Jones, 2003), studies which help cast comparative light on Johnston and Williams' British survey.

Two longer works should certainly stand as markers in their respective fields. In his *Geography and history*, Alan Baker (2003) documents the connections between these fields from the 'middle ground' of historical geography, in which field he has had a major influence. He is also motivated by the concern to see historical geography become once again so central to human geography that, through its very hybridity, it effectively extinguishes itself as a subdisciplinary concern, at least in the ways it has been hitherto considered. Where Baker looks at geography and history as intellectual concerns principally in the work of North American, French and British historians and geographers, Neil Smith takes as his focus the overlapping worlds of politics and geography in the life and work of Isaiah Bowman (Smith, 2003). Smith's argument coheres around the ways in which the American political geographer Isaiah Bowman was the architect of that period of

global dominance by the United States now termed the 'American Century', beginning in 1898 and ending – well, has it? How may we understand his book? It is 'a history of geography, but even more a geography of history' (Smith, 2003: xvii). It is a biography, but not a standard one in as much as we are told as much about the 'lost geography' of the American Century – and the later historiographical invisibility of Bowman – as we are of his pre-eminence as the man who counselled Wilson at Versailles and after. It is a history of geography's books: Bowman's *The new world* (1919) became a sort of 'handbook for the budding American Century' (Smith, 2003: 26). It is a thoroughly researched account of the American geopolitical imagination, of the ideas behind the notion of an American Lebensraum and 'the specific and global historical geography of the U.S. ascension to power' (Smith, 2003: 319). In one sense, it may be seen as a coincident expression of Livingstone's call for attention to the significance of space in writing 'geographical biography' (Livingstone, 2003a: 182–83). It is in another sense a quite superb illustration of how central the history of geography remains (and, *pace* Baker and Bowman, the geography behind history), to questions of geography's influence upon political history.

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